

Long & Short-time User Memory with Image selection

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CONTENTS

.....1
CONTENTS 2
1 Project overview 3
2 Project objective 4
RQ1 Memory reflection and user control 4
RQ2 Taste expression and specification 4
3 System design (including low-fidelity prototype) 5
3.1 System Overview Specifying fashion preferences through memory and interaction 5
3.2 Input-to-Memory Model and Memory Layer Input interpretation and memory management 5
3.3 Context-Aware Memory Retrieval Retrieving memory relevant to the current request 6
3.4 Item Feature Map Construction Converting the current request and memory into target attributes 6
3.5 Item Retrieval and Final Suggestion Comparing target attributes with actual products 7
3.6 Feedback and Memory Update Interpreting selection reasons and updating memory 7
3.7 UI Design 8
4 User study design 10
4.1 Purpose and hypotheses 10
4.2 Participants and experimental design 10
4.3 Procedure 11
4.4 Measures 12
5 Statistical analysis plan 14
5.1 Variables and unit of analysis 14
5.2 Statistical tests 15
5.3 Order effect check, exploratory analysis, and interpretation criteria 15
6 Future work and next steps 16
7 Appendix 17

1 Project overview

This study targets users who want to find fashion items or outfits that fit both their own taste and their current situation. Even when users can recognize and pick clothes they like, they may find it hard to describe the fitness, color, or mood they want from the start in concrete fashion times or text conditions. In addition, the style a user usually prefers, and the conditions required by the current situation can differ. For example, a user who usually likes relaxed-fit pants may want slim pants for an upcoming presentation.

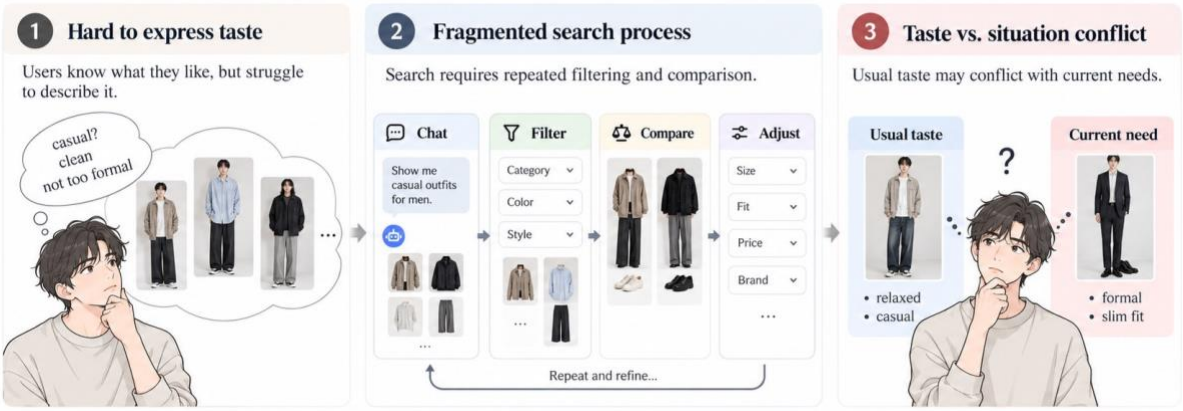


Figure 1. Problem illustration. Users face difficulty expressing fashion taste, repeated. search refinement, and conflicts between usual preferences and situational needs.

As the comparison condition, this study uses the current practice of deciding a recommendation direction through conversation with a chatbot and then finding products with the search and filter functions of a shopping service. In this process, users must translate the feeling they want into searchable conditions, compare the products shown, and adjust the conditions. When their usual taste differs from the requirements of the current request, they must also convey which information to keep and how far it should apply. This study focuses on support that reduces this burden of expression and adjustment. Whether a situational choice is generalized into a lasting preference, or whether users must re-explain their existing taste, are issues to be examined; we do not assume that existing tools in general fail to support them.

To this end, we propose LUMI (Long&Short-time User Memory with Image-selection), a conversational fashion recommendation agent that combines long and short-time memory with image-selection-based interaction. The user enters the desired style or conditions as text and selects a look or item they like from the image candidates presented. Based on the user's selections and conversation, the system forms interpretive hypotheses about the reasons for the selection and builds a target Item Feature Map for product retrieval. When an unclear interpretation would affect the next recommendation, the system offers a short confirmation question or a set of comparison images, and it refines the recommendation conditions while distinguishing confirmed preferences from tentative interpretations.

Long-time memory preserves preferences that the user has confirmed and agreed to reuse in later requests. Short-time memory handles the requirements and situation of the current request, the user's selections and modifications, and interpretations still being confirmed. The system presents candidates using the memory relevant to the current request and adjusts recommendations according to feedback. It may suggest wrapping up the exploration, but the user decides the final outfit and whether to end the exploration. Whether a newly confirmed preference is used in the long time is also subject to the user's separate consent.

2 Project objective

The goal of this project is to help users specify the conditions they want through image selection and conversation without describing their taste in detail from the start, and to find fashion items that reflect both their usual taste and the requirements of the current situation. To this end, we design LUMI, which combines long and short-time memory with selective confirmation questions, and evaluate users' sense of control, outfit appropriateness, and the experience of expressing taste.

- **Support for taste exploration**: help users gradually specify the style they want through image candidates and feedback.
- **Separation of long & short-time memory**: distinguish lasting preferences the user has agreed to reuse from the requirements and feedback of the current request.
- **Preference management**: do not arbitrarily fix tastes or attributes that were not provided and distinguish confirmed preferences from tentative hypotheses proposed by the system.
- **Guarantee of user decision-making**: information to be stored in long-time memory and reused later requires the user's separate consent, and the user decides the final selection and whether to end the exploration.

This goal is examined through two research questions. RQ1 evaluates sense of control and outfit appropriateness in a usage flow that supports the separation of long- and short-time memory and the review and revision of information. RQ2 evaluates expression burden, style specification, and the feeling of being understood in an interaction that includes image selection and selective confirmation questions. The two questions separate the evaluation focus; they do not imply an experiment that isolates the independent effect of each feature.

RQ1 Memory reflection and user control

Compared with the Baseline, does LUMI, which manages long and short-time memory separately and lets users review and revise how their information is reflected, increase users' sense of control and the appropriateness of the final outfit (taste reflection and situational fit) over the course of two requests within one session?

- **Memory fidelity**: retention of existing taste and reflection of newly confirmed preferences are examined through LUMI-only evaluation. Prevention of over-generalizing temporary requirements is left as a follow-up evaluation item.
- **Sense of control**: the degree to which users feel they understand what information is reflected in recommendations and can revise it in the direction they want.
- **Outfit appropriateness**: the degree to which the final outfit matches the user's taste and the conditions of the current request.

RQ2 Taste expression and specification

Compared with the Baseline, does LUMI, which combines image candidate selection, selective confirmation questions, and free text input, reduce the burden of conveying taste and requirements, and increase the specification of the desired style and the feeling of being understood by the system?

- **Expression burden**: the effort needed to find precise fashion times or write long descriptions.
- **Style specification**: the degree to which the desired style and selection criteria become clear through interaction.
- **Feeling of being understood**: the degree to which users feel the system correctly grasps their preferences and intentions.

3 System design (including low-fidelity prototype)

3.1 System Overview

Specifying fashion preferences through memory and interaction

We propose LUMI, a memory-based fashion recommendation system that reflects both the user's lasting taste and the requirements of the current request. LUMI is designed to gradually specify recommendation conditions using previously confirmed preferences and feedback obtained from conversation and image selection, so that the user does not have to describe every feature of the desired clothing from the start.

Figure 1 shows the overall system flow of LUMI. The system interprets preferences and their scope of application from user input and manages them as short- and long-time memory (3.2), then retrieves the memory relevant to the current request (3.3). Based on the current request and the retrieved memory, it builds the target attributes of the product to be found as an Item Feature Map (3.4) and presents recommendation candidates by comparing them with the attributes of actual products (3.5). The user's selections and modifications are reflected in the conditions of the current request and the next recommendation, and preferences the user has agreed to reuse are reflected in long-time memory (3.6).

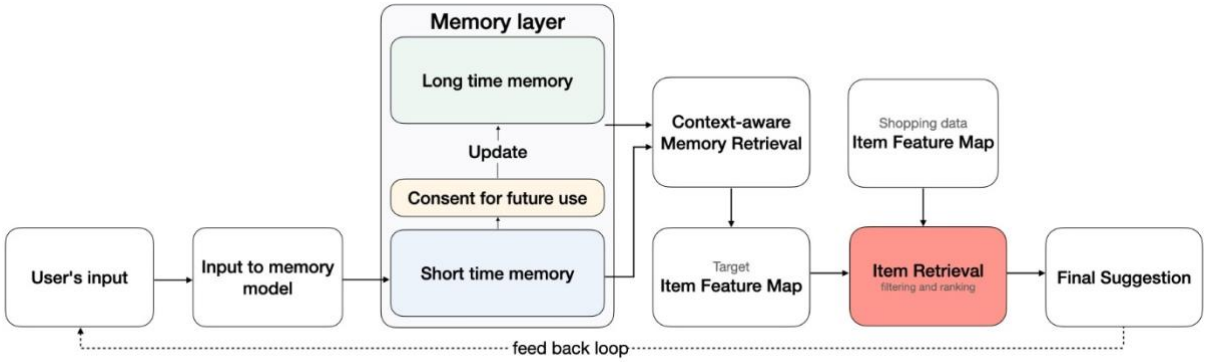


Figure 2. Memory-based fashion recommendation flow of LUMI. User input is organized by the content of the preference and its scope of application. Memory relevant to the current request is used to build the target attributes of the Item Feature Map, which are compared with actual products to produce recommendations. Selections and modifications of the recommendation results update the conditions of the current request, and preferences the user has agreed to reuse are reflected in long-time memory. Solid lines indicate information transfer and processing flow; dotted lines indicate the loop of user feedback.

3.2 Input-to-Memory Model and Memory Layer

Input interpretation and memory management

The **Input-to-Memory Model** is the stage that extracts preferences and requirements from the user's utterances and image selections and organizes them into content to be remembered. Each item is linked with which garment and situation it concerns, whether it was expressed directly by the user or inferred by the system, and which utterance or selection it is based on. For example, the input "I like tops loose but pants slim" is organized into separate preferences for tops and pants. For "I want to wear slim pants for this presentation," the input records the fit of the pants together with its scope of application, namely a requirement limited to this presentation.

The LLM extracts candidate content to remember from the input and proposes interpretations or confirmation questions for parts whose meaning is unclear. When the user says, "I like a vintage feel," the system preserves the expression and the conversational context but does not immediately fix it as a preference for a specific color or material. The meaning is later specified through the user's explanations or image selections, and interpretations inferred by the system are managed separately from content confirmed by the user.

The Memory Layer manages this information as Long-time Memory and Short-time Memory. Long-time memory handles preferences the user has confirmed and agreed to reuse in later requests. Short-time memory handles the requirements of the current request, the wearing situation, selections and modifications, and interpretations under confirmation. Therefore, the fact that a specific item was selected in the current request, or that the same selection was repeated, is not by itself stored as a lasting taste.

3.3 Context-Aware Memory Retrieval
Retrieving memory relevant to the current request

Context-Aware Memory Retrieval is the stage that selects, from stored memory, the content to consult for the current recommendation. Based on the type of garment currently being sought and the wearing situation, the system retrieves the requirements held in short-time memory and the relevant preferences in long-time memory. For example, for a request to find pants for a presentation, it consults both the conditions for pants confirmed in the current conversation and the user's usual preferences for the color and fit of pants.

During retrieval, the system checks not only the content of each memory but also its scope of application. A condition wanted only for a specific presentation is not applied as is to a request for everyday clothes, and the short-time memory of a previous request is not automatically carried over to a new request. In contrast, selections and modifications that continue within the same request are still used as grounds for understanding the current conditions.

The result of this stage is the set of memories relevant to the current request together with their grounds. Each memory is linked with the target garment and situation, whether it is a requirement of the current request or a lasting preference, and the original utterance or selection it is based on. In the next stage, this information is combined with the current request to build the target attributes used for product retrieval.

3.4 Item Feature Map Construction
Converting the current request and memory into target attributes

The **Item Feature Map** represents the features of clothing as common attributes such as type, color, and fit. The system interprets the current request and the retrieved memory to build the target Item Feature Map of the product to be found this time. For each attribute, it records the value and the grounds for the judgment and distinguishes conditions that must be satisfied from optional preferences. It also distinguishes values the user stated or confirmed directly from values inferred by the system, and attributes with insufficient information are left undetermined.

When the current request conflicts with existing long-time memory, the requirement the user stated explicitly takes priority for that request. For example, if a user who usually prefers navy relaxed-fit pants requests "I want slim pants for this presentation," the target attributes are built as "type: pants, fit: slim, color: navy." The relaxed fit, which conflicts with the current requirement, is excluded from the target attributes, while the navy color preference, which does not conflict, is retained. This adjustment applies only to the conditions of the current retrieval and does not change existing long-time memory.

Attributes inferred from context are distinguished from the user's requirements. A neat style may be inferred from the presentation situation, but a specific material or design is not fixed as a required condition. When a core requirement is hard to interpret, or the priority among conflicting conditions cannot be datamined, the system confirms with the user before building the target attributes.

3.5 Item Retrieval and Final Suggestion
Comparing target attributes with actual products

Item Retrieval is the stage that finds recommendation candidates by comparing the target Item Feature Map with the attributes of actual products. Product data are collected by crawling and labeled with the same criteria as the target attributes to form a per-product Item Feature Map. The target Map represents the conditions of the product being sought, and the per-product Map represents the features confirmed for that product.

Retrieval consists of candidate selection and ranking. Products that satisfy the required conditions are selected first, and within that set the presentation order is detimeined by the degree of match with optional preferences. Target attributes inferred by the system may be used to adjust the ranking but do not replace the required conditions stated by the user. When a product's attribute has not been confirmed, the product is not judged to satisfy that condition.

Recommendation candidates are presented together with product images. If no product satisfies the required conditions, the system reports that there are no results and asks the user which conditions to adjust. The user can select a presented candidate or request further exploration, and once a selection is confirmed, the result is organized as the final suggestion.

3.6 Feedback and Memory Update
Interpreting selection reasons and updating memory

The user can select all or part of a recommended item or convey modifications as text. The system distinguishes the target and the attribute that the feedback refers to. For example, the response "I like the color of these pants, but I'd prefer a more relaxed fit" is organized as a positive evaluation of the color of those pants and a modification request for the fit.

Selecting an image does not by itself mean the user prefers every attribute in it. The LLM forms hypotheses about the reason for the selection by referring to the attribute differences between the selected candidate and the other candidates, existing memory, and the current request. Unselected candidates are likewise not immediately recorded as disliked.

When the conditions of the next recommendation would change depending on the hypothesis, the system confirms the reason for the selection with a short question or an image comparison. If the user finds it hard to explain the reason, only the fact of the selection is recorded, and the inferred reason is kept unconfirmed. Confirmation questions are therefore not repeated after every selection but used to obtain the information needed for the next recommendation.

Confirmed selection reasons and modification requests are reflected in short-time memory. The system rebuilds the target Item Feature Map based on the updated memory and repeats product retrieval. During this process, the user can confirm the desired item or stop the exploration.

Long-time memory is updated separately from the adjustment of the current request. The system presents the preferences to be reused later to the user and stores the content the user has consented to. When replacing an existing preference, it confirms whether the change is an exception limited to the current request or a lasting change in taste. Confirming a recommendation result or repeating the same selection does not by itself change long-time memory automatically.

3.7 UI Design

The web UI of LUMI is designed so that users can adjust the conditions they want through conversation while browsing image candidates. The exploration screen places the candidate image and selection area on the left and the conversation area on the right, with items at the top for starting a new request and moving to saved preferences. Instead of exposing the entire internal Item Feature Map on the main screen, the UI presents the guidance and choices needed for the current action, and detailed information about what has been reflected can be opened when needed. The core flow of the web UI design is as follows. Web UI Design

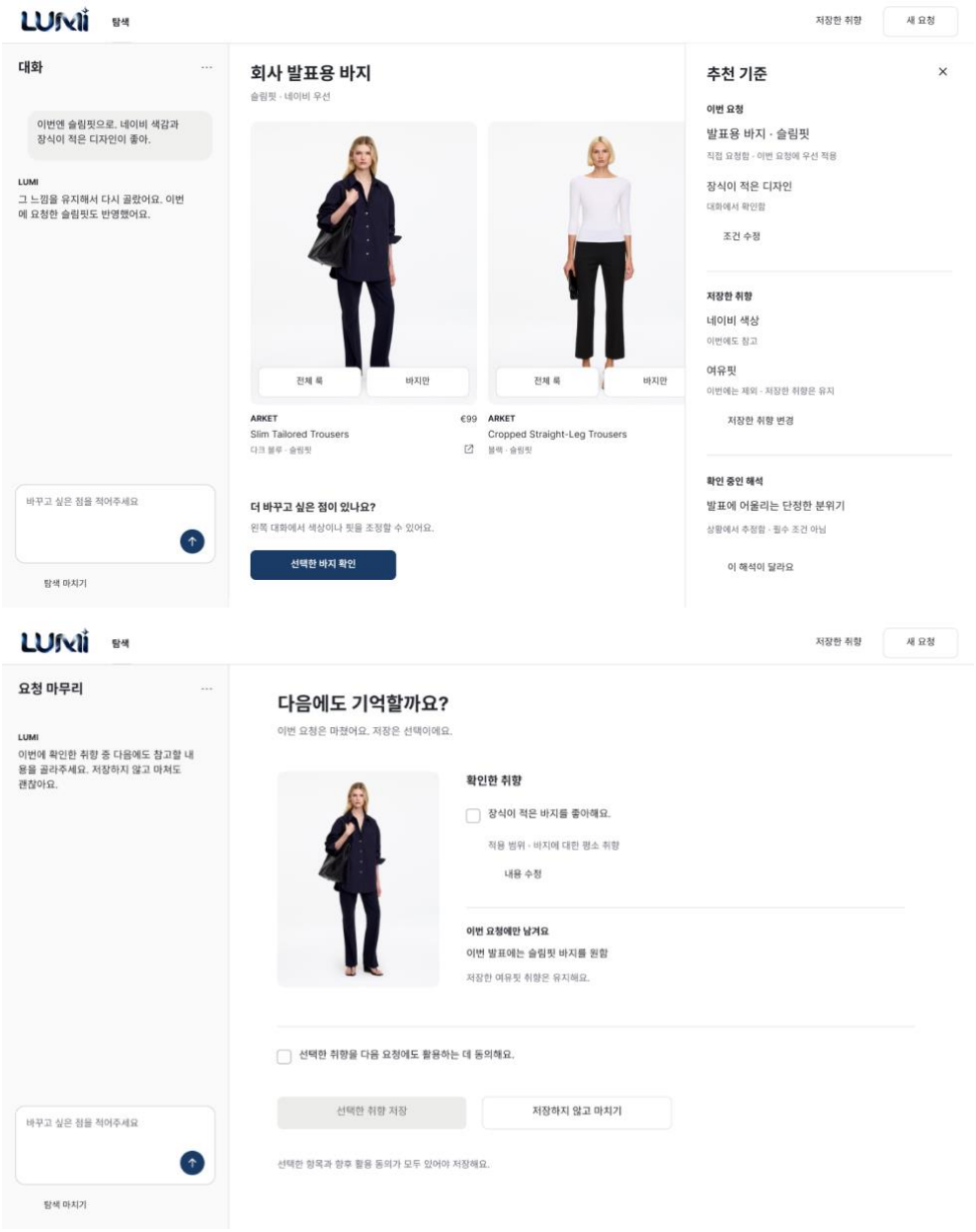


Figure 3. Image exploration screen of the LUMI web UI. The user selects a whole look or a specific item on the left and supplements the request in the conversation area on the right. The number of candidates and the clothing shapes are an example of the screen layout.

Initial preference registration and new requests. On first use, users can convey their usual taste through image selection or a short description. After checking and revising the content organized by the system, they either consent to its future use and save it, or start a request right away without saving. On the new request screen, users enter the garment they need and the wearing situation and check the related saved preferences together. For example, even if a user usually prefers navy relaxed-fit pants, a request such as “I want slim pants for this company presentation” makes the current recommendation prioritize a slim fit while referring to the navy color preference. The existing relaxed-fit preference is kept as it is.

Image exploration and partial selection. Recommendation candidates are presented as image cards that can be compared. Users can choose several candidates, or select a specific target within one image, such as the whole look or only the pants. Even when it is hard to describe the desired features precisely in words, exploration can continue with image selection as the starting point. “None of these appeals to me” and free text input are also provided so that selection is not forced. A selection is treated as feedback on the selected target; the system does not conclude that the user prefers every attribute of the image or dislikes the unselected candidates.

Confirming selection reasons when needed. When the interpretation of a selection reason would affect the next recommendation, the system shows the selected image together with a short question such as “Did you like the navy color and the minimal design?” Users can answer with “Yes,” “Only the color,” “Something else,” or “Not sure,” or explain in their own words. When a difference that is hard to put into words matters for the recommendation, an image comparison is provided. Questions and comparisons can be skipped; if the reason cannot be confirmed, only the fact of the selection is kept and the inferred reason is not fixed.

Re-recommendation and checking or revising reflected information. Confirmed selection reasons and modification requests are applied to the current request to update the candidates, and the user repeats image selection and conversation. Opening “View reflected information” shows, in the right panel, “This request,” “Usual preferences referred to,” and “Interpretations still being confirmed” as separate groups. In the example, the slim fit for the presentation is shown as a condition of this request, navy as a referenced preference, and the relaxed fit as a preference excluded this time but preserved. Users can revise the conditions of this request or indicate that the system’s interpretation is wrong; changes to saved preferences are handled separately.

Final selection and long-time memory storage. The wrap-up screen summarizes the selected item and the conditions reflected and offers “Confirm this item” and “Keep exploring.” The system may suggest wrapping up, but the final decision is made by the user. The memory storage screen then shows the specific preferences the user confirmed and their scope of application and asks separately which items to save and whether they consent to future use. Items and consent are unchecked by default, and “Finish without saving” is also provided. For example, the confirmed preference “likes pants with minimal detailing” can become a candidate for storage, whereas the slim-fit requirement for this presentation remains limited to this request. Confirming an item or answering a selection-reason question does not by itself update long-time memory.

Next request and exception flows. In a new request for weekend pants, the system refers to the relevant long-time memory but does not automatically carry over the slim fit required for the earlier presentation. “Minimal design” is used in the next request only if the user consented to it in the earlier storage step. When no product satisfies the required conditions, the system reports that there are no results and lets the user decide whether to change the conditions, keep them, or end the exploration. Users can also end the exploration without selecting an item. Through this flow, the adjustment of the current recommendation, the final item selection, and the storage of memory for later requests are organized as separate user decisions.

4 User study design

Participant Survey Link

4.1 Purpose and hypotheses

This user study tests RQ1 and RQ2. The sub-criteria of each RQ are used as the measured constructs, and for each criterion we include paired items compared against the Baseline and LUMI-only items evaluated only for LUMI.

H1 (RQ1). LUMI, which manages long- and short-time memory separately and lets users review and revise how their information is reflected, is rated higher than the Baseline on (a) memory fidelity, (b) sense of control, and (c) appropriateness of the final outfit. Because memory fidelity has no counterpart in the Baseline, it is examined through LUMI-only evaluation.

H2 (RQ2). LUMI, which combines image candidate selection, selective confirmation questions, and free text input, is rated lower than the Baseline on (a) expression burden and higher on (b) style specification and (c) the feeling of being understood.

SUS and NASA-TLX are supplementary measures that describe usability and workload, independent of the hypotheses.

4.2 Participants and experimental design

We recruit 20 to 24 adults who have purchased fashion products online. The study uses a within-subject design (repeated measures). The independent variable (IV) is the system (Baseline vs LUMI, two levels), and the dependent variables (DV) are the questionnaire scores for each sub-criterion in 4.4. All participants perform the same type of task with both systems and evaluate each condition immediately after performing it.

Because individual differences such as fashion knowledge and the ability to express preferences are large, we chose a within-subject design, which cancels out individual differences by comparing the two scores of the same person. In a within-subject design, the order effect, in which the condition performed first influences the evaluation of the later condition, and the carryover effect, in which the scenario or know-how learned in the first task carries over to the second task, can act as confounding variables. To control for them, the order of conditions is counterbalanced.

Half of the participants perform Baseline → LUMI and the other half LUMI → Baseline. Because repeating the same scenario in both conditions would increase the carryover effect, we prepare two outfit scenarios matched in scope and difficulty (A: "an outfit for a blind date," B: "an outfit for a company presentation") and counterbalance the scenario–condition assignment. This yields four sequences, order (2) × scenario assignment (2), and participants are assigned to the four sequences in rotation by participant number. Order and scenario are treated as control variables in the analysis, and whether an order effect occurred is checked in 5.3.

The Baseline is fixed as a combination of tools users can use today: a general-purpose LLM chatbot (ChatGPT) and a fashion shopping app (Musinsa). Participants type their request into ChatGPT to obtain a recommendation direction (style, items, search keywords) and then use search and filters on Musinsa to find actual products and finalize an outfit. The Baseline therefore also includes conversation and product exploration, but it lacks the separation of long- and short-time memory, image candidate selection, and confirmation questions about selection reasons. To keep the conditions equal, ChatGPT is used with memory and personalization turned off, in a new conversation for each participant, without image generation or web search, and Musinsa is used in a logged-out state with search and filters only, without personalized recommendations.

Because tool differences would be confounded with the system effect if participants used different tools, this combination is applied identically to all participants, and the tools participants usually use are recorded only as prior experience in pre-survey item Q (2-4). Before the main experiment, a pilot with about three participants checks scenario difficulty, item comprehension, and session duration.

4.3 Procedure

The full procedure runs in the following order and is expected to take 50 to 60 minutes per participant. Condition 1 and Condition 2 are either Baseline or LUMI depending on the sequence assigned to the participant. The questionnaire is administered with Google Forms in two versions matched to the condition order. The Google Form link submitted with this report is for participants performing Baseline to LUMI, in which Section 6 (Baseline evaluation) precedes Section 7 (LUMI evaluation). Participants performing LUMI → Baseline use a separate Google Form in which only the positions of Sections 6 and 7 are swapped. Because the two forms have identical item numbers and wording and differ only in section order, responses are merged by item number for analysis.

- Study briefing and consent Q(1-1). The purpose of the study, the duration, the fact that both systems will be used, and the right to withdraw are explained, and consent is obtained. The facilitator checks the sequence (condition order and scenario assignment) by participant number.
- Pre-survey Q(2)–Q(5). Participants answer items on basic information, fashion knowledge and the ability to express preferences, situation-dependent selection tendencies, and situations in which AI recommendation is needed.
- Condition 1 task. Participants perform the task with the assigned system. The task structure is the same for both systems and consists of two requests so that the process in which long-time memory is formed and applied to the next request can be observed within one session. In the LUMI condition, a short preference registration (onboarding) is conducted right before the task. Participants select outfit images close to their usual preferences and write one or two sentences about the style they usually wear; once they confirm the content organized by the system, it is stored in long-time memory and serves as the reference point for "existing taste."

Request 1, free request. Participants freely request, in their own words, clothes they want to find now (e.g., "a fall outer layer I could wear these days"). Because no situational condition is given, the selections and modifications in this request reflect the participant's usual taste. In LUMI, participants choose at the end of the request whether to store the confirmed preferences in long-time memory, and this is the point at which long-time memory accumulates.

Request 2, situational request. The facilitator presents the assigned scenario (A: "an outfit for a blind date," B: "an outfit for a company presentation"), and participants enter it and explore until they finalize an outfit they like. Because this request creates a conflict between usual taste and situational requirements, we observe whether the taste stored during onboarding and Request 1 is retained while the situational condition is applied to this request. At the end, participants again choose whether to store the preferences revealed this time as usual taste or to use them only for this request.

In the Baseline condition, participants perform the same two requests, but for each request they first discuss the recommendation direction with ChatGPT and then find products on Musinsa to finalize an outfit. Because the two requests continue within the same ChatGPT conversation, how the taste revealed in Request 1 is reflected in the recommendations and searches of Request 2 can be contrasted with LUMI.

- Condition 1 evaluation. Immediately after the task, participants complete the paper NASA-TLX and then answer the paired items for that condition (Q(6) for Baseline, Q(7) for LUMI). NASA-TLX comes first because the workload rating depends on memory immediately after the task.
- Condition 2 task. Participants perform the same structure as the Condition 1 task (Requests 1–2) with the remaining system, using the remaining scenario for Request 2. A 2–3-minute break is given between conditions.
- Condition 2 evaluation. As in the Condition 1 evaluation, participants complete the paper NASA-TLX and the paired items for that condition.
- Post-survey Q(8)–Q(10). After both conditions, participants answer the SUS for LUMI, the LUMI-only items (preference and situation reflection, interaction), and the open-ended questions.
- Debriefing. The facilitator spends 2–3 minutes going over the moments mentioned in the open-ended answers and what participants felt about the differences between the two systems.

4.4 Measures

Sub-criteria and items. The table below links the sub-criteria of each RQ to the questionnaire items. Q(6) and Q(7) are paired items with identical wording, answered as nine items each immediately after the Baseline and LUMI tasks. Because the Baseline tool is fixed as a general-purpose LLM chatbot, Q(6-10) (recording the recalled tool) is removed. Q(9-6), Q(9-7), Q(9-9), Q(9-13), and Q(9-18) are optional items that participants who did not encounter the relevant situation may skip, and non-responses are treated as "not applicable."

RQ	Sub-criterion	Baseline ↔ LUMI paired items	LUMI-only items
RQ1	Memory fidelity	–	Q(9-2) taste–situation separation reflected, Q(9-14) existing taste retained, Q(9-18) stored preferences reflected
RQ1	Sense of control	Q(6-3)↔Q(7-3) intervention in process and result, Q(6-6)↔Q(7-6) understanding and revising reflected information	Q(9-3) choice of storage scope, Q(9-4) right to end, Q(9-8) comprehensibility of interpretation
RQ1	Outfit appropriateness	Q(6-1)↔Q(7-1) taste reflection, Q(6-2)↔Q(7-2) situational fit	–
RQ2	Expression burden	Q(6-7)↔Q(7-7) conveying without times or long descriptions	Q(9-5) conveying by selection alone, Q(9-7) one-tap responses, Q(9-10) questions only when needed, Q(9-13) proceeding without explaining

RQ	Sub-criterion	Baseline ↔ LUMI paired items	LUMI-only items
RQ2	Style specification	Q(6-4)↔Q(7-4) style specification, Q(6-8)↔Q(7-8) clarification of selection criteria	Q(9-1) recommendations narrow with each selection, Q(9-19) image selection helps organize style
RQ2	Feeling of being understood	Q(6-5)↔Q(7-5) system understands, Q(6-9)↔Q(7-9) accurate reflection of conveyed intent	Q(9-6) selection-reason hypothesis hit, Q(9-9) timing of the confirmation suggestion, Q(9-20) previous selection reflected
Manipulation check	Validity of taste registration and scenario	–	Q(9-16) onboarding reflects usual taste, Q(9-17) situation requires choices different from usual

Q(9-16) and Q(9-17) are manipulation checks rather than hypothesis-testing items. They confirm whether the taste stored during onboarding and Request 1 reflected the participant's actual taste and whether the scenario in Request 2 created a taste–situation conflict, and thus whether the premise for interpreting the RQ1 results held. "The second request" in Q(9-18) and "previous selection" in Q(9-20) refer to the flow from Request 1 to Request 2. Q(9-15) (non-carryover of temporary requirements) can only be answered if another request follows the situational request; because the task ends at Request 2 (situational request), it is not measured in this study and is excluded from the form.

Pre-survey. Basic information and experience with AI recommendation Q(2-1)–Q(2-5), preference awareness and articulation ability Q(3-1)–Q(3-4), preference for image-based judgment Q(3-5), consideration of situations Q(4-1)–Q(4-2), taste–situation conflict Q(4-3)–Q(4-4), reluctance to have situation-specific search history reflected in usual recommendations Q(4-5), and situations in which AI help is needed Q(5-1)–Q(5-2). Q(4-3)–Q(4-5) are descriptive data supporting the design motivation of RQ1, and Q(3-1)–Q(3-5) and Q(4-3)–Q(4-4) are exploratory moderating variables.

Standard scales and open-ended questions. Raw NASA-TLX is measured on paper with six subscales (0–100; the performance item is reverse-coded as 100 − response) immediately after the task in both conditions. SUS Q(8-1)–Q(8-10) (0–100) is measured for LUMI. The two systems are introduced to participants only as "System A" and "System B," and which one is the research team's proposed system is not revealed until debriefing. Open-ended questions are Q(10-1)–Q(10-6). Q(9-11)–Q(9-12) (exploration and discovery) fall outside the RQs and are reported only as supplementary results.

5 Statistical analysis plan

5.1 Variables and unit of analysis

The independent variable is the system (Baseline vs LUMI, two within-subject levels). Condition order (Baseline first / LUMI first) and scenario assignment (A/B) are variables controlled through counterbalancing and are used only for the order effect check (5.3). The dependent variables are (1) the paired items measured repeatedly in both conditions, (2) the LUMI-only items, and (3) NASA-TLX (both conditions) and SUS (LUMI). Individual Likert items are treated as ordinal data. Considering the statistical power of the small sample (n≈20) and the burden of multiple comparisons, category scores (item means, 1–5) at the level of each sub-criterion are used as the main unit of analysis. Non-responses to optional items are treated as missing. A category score is computed as the mean of the answered items only when at least half of the items in that category were answered; otherwise, the participant is excluded from that test, and the n for each test is reported.

RQ	Category	Items	Comparison
RQ1	Outfit appropriateness	Q(6-1), Q(6-2) ↔ Q(7-1), Q(7-2)	Baseline
RQ1	Sense of control	Q(6-3), Q(6-6) ↔ Q(7-3), Q(7-6)	Baseline
RQ1	Memory fidelity (LUMI-only)	Q(9-2), Q(9-14), Q(9-18)	Neutral point 3
RQ1	Sense of control (LUMI-only)	Q(9-3), Q(9-4), Q(9-8)	Neutral point 3
RQ2	Expression burden	Q(6-7) ↔ Q(7-7) (single item)	Baseline
RQ2	Style specification	Q(6-4), Q(6-8) ↔ Q(7-4), Q(7-8)	Baseline
RQ2	Feeling of being understood	Q(6-5), Q(6-9) ↔ Q(7-5), Q(7-9)	Baseline
RQ2	Expression burden (LUMI-only)	Q(9-5), Q(9-7), Q(9-10), Q(9-13)	Neutral point 3
RQ2	Specification (LUMI-only)	Q(9-1), Q(9-19)	Neutral point 3
RQ2	Being understood (LUMI-only)	Q(9-6), Q(9-9), Q(9-20)	Neutral point 3
Supplementary	NASA-TLX	Paper, 6 subscales, both conditions	Baseline
Supplementary	SUS	Q(8-1)–Q(8-10)	68

Internal consistency is checked before combining items (Cronbach's α for three or more items, Spearman-Brown for two items; ≥.70 good, .60–.70 acceptable for exploratory purposes; paired categories are computed separately for Baseline and LUMI). Even if the criterion is not met, the main tests are performed on the pre-defined category scores, and item-level results are presented as supplementary results not included in the Holm correction. Style specification and the feeling of being understood are separate sub-criteria of H2 and are therefore tested separately. SUS and NASA-TLX are standard scales and are scored with their original methods.

5.2 Statistical tests

Due to the Baseline–LUMI comparison uses data from the same participants measured repeatedly in two conditions; paired tests are used. Since the sample is small and the scores are Likert-based, the Wilcoxon signed-rank test is the main test for comparing category scores and the single item (expression burden), and the paired t-test is reported as a sensitivity analysis. Because the two scores of the same participant are correlated, the independent-samples t-test is not used.

Because the LUMI-only categories have no comparison condition, a one-sample Wilcoxon signed-rank test against the neutral point of 3 is performed, and a one-sample t-test is reported as a sensitivity analysis. Cases with a difference of zero are excluded from the Wilcoxon test (Wilcoxon method), and the number of excluded participants is reported for each test.

Since NASA-TLX is measured in both conditions, the total score (mean of the six subscales) is compared with the Wilcoxon signed-rank test, and the subscales are reported as descriptive statistics by condition. SUS is compared with the industry benchmark of 68 using a one-sample Wilcoxon signed-rank test, with a one-sample t-test reported as a sensitivity analysis.

The pre-survey is reported as descriptive statistics (M, SD, 95% CI). Because there are only two conditions, ANOVA and the Friedman test are not used. All tests are two-sided with $\alpha=.05$, and the Holm method is applied for multiple-comparison correction within each RQ (RQ1: 2 Baseline comparisons + 2 LUMI-only; RQ2: 3 Baseline comparisons + 3 LUMI-only). For effect sizes, the matched-pairs rank-biserial correlation (r_{rb}; .1/.3/.5) is reported for the Wilcoxon signed-rank test, and d_z (mean of difference scores / standard deviation of difference scores; .2/.5/.8) for the t-test.

5.3 Order effect check, exploratory analysis, and interpretation criteria

Order effect check. Although counterbalancing distributes order effects evenly across the two conditions, we check whether order influenced the results. The difference score for each category (Δ = LUMI − Baseline) is compared between the Baseline-first group and the LUMI-first group using the Mann-Whitney U test. Because the two groups consist of different participants, an independent-samples test is used. If the difference is significant, an order effect is reported, and results are presented by group. The same check is applied to scenario assignment (A→LUMI / B→LUMI). Because this check is exploratory, no multiple-comparison correction is applied.

Exploratory analysis. The relationship between individual differences in the pre-survey and Δ is explored with Spearman correlation. The preference articulation score is the mean of Q(3-1)–Q(3-4), the taste–situation conflict score is the mean of Q(4-3)–Q(4-4), and preference for image-based judgment Q(3-5) is analyzed separately as a single item. We expect that the lower the preference articulation ability, the larger the Δ in the RQ2 categories, and the greater the taste–situation conflict, the larger the Δ in the RQ1 categories. The relationship between the task-level degree of conflict Q(9-17) and the RQ1 category Δ is checked in the same way.

For the accuracy of onboarding registration Q(9-16), the correlation with Q(7-1) and Q(9-14) is checked, and a sensitivity analysis excluding participants who responded 2 or lower is also reported. Experience with AI recommendation is split by the response to Q(2-4) into 0 times (never used) and 1 or more times, and differences in SUS and Δ between the two groups are checked with the Mann-Whitney U test. If either group has fewer than 5 participants, only descriptive statistics are reported without a test.

Because the correlations and group comparisons in 5.3 are exploratory, no multiple-comparison correction is applied, and Spearman ρ is reported with 95% confidence intervals. Open-ended responses Q(10-1)–Q(10-6) are inductively coded independently by two researchers, who then agree on a codebook and resolve disagreements through discussion. The coding results support the interpretation of the quantitative results; in particular, Q(10-2) (experience of taste–situation separation) is used to interpret memory fidelity, Q(10-3) (intentions outside the buttons) expression burden, and

Q(10-5) (hypothesis errors, timing of the suggestion) the feeling of being understood. Differences between the two systems that participants mention during debriefing are organized with the same codebook.

Interpretation criteria. A hypothesis is considered supported for a given sub-criterion if $p<.05$ after Holm correction and the direction of the effect matches the hypothesis. Non-significant results are reported as "insufficient evidence of a difference," together with the effect size and the possibility of a ceiling effect. The one-sample test results for the LUMI-only items are interpreted as descriptive evidence of whether participants generally agreed with the experience in question, not as evidence of superiority over the Baseline. If the manipulation checks Q(9-16) and Q(9-17) are below the neutral point, this limitation is stated in the interpretation of the RQ1 results.

6 Future work and next steps

In the future work, we plan to implement a working prototype based on the system architecture of LUMI designed here and to verify the effectiveness and usability of the system through a suer study. First, based on the Feature Map we have built, we will validate VLM-based clothing feature extraction models and design the product database and the personalized recommendation engine. We also plan to implement the long- and short-time Memory Layer that integrates the user's long-time fashion taste with the requirements of the current session.

We will then complete an adaptive feedback system that uses this Memory Layer together with an LLM. The LLM will analyze the features of the outfits the user selects to infer the reasons for their preferences, and when further confirmation is needed, the system will ask the user for brief feedback. The collected feedback will be reflected in the preferences of the current session to gradually improve the recommendation results; the user will be able to make a final selection at any time, and the agent will suggest ending the session once sufficient interaction has taken place.

Once the core features are implemented, we will develop a high-fidelity prototype that integrates the front end and back end and then conduct a user study comparing LUMI with existing fashion recommendation systems to evaluate the burden of expressing taste, recommendation satisfaction, the effect of taste discovery, and interaction efficiency, both quantitatively and qualitatively.

Finally, based on the results of the user study, we plan to refine the preference inference and feedback strategies. Using the usage data collected from real user experience, we aim to improve the accuracy of the LLM's inference of selection reasons and to develop the system in a direction that minimizes unnecessary questions by considering the uncertainty of user preferences and its effect on recommendation results. Furthermore, by effectively distinguishing and reflecting long-time taste and temporary preference changes across multiple sessions, we intend to continuously improve the accuracy of personalized recommendation and the user experience.

7 Appendix

The questionnaire items below are reproduced verbatim from the Google Form administered to participants. Because the study is conducted with Korean-speaking participants, the items are kept in the original Korean to preserve the exact wording they will see. Item numbers Q(section-item) match those cited in Sections 4 and 5.

User study Questionaire(In Korean)

섹션 1. 연구 참여 동의

- **Q(1-1)** 연구 안내문을 읽고 연구 참여 및 응답 수집에 동의하십니까?
    - ○ 동의함
    - ○ 동의하지 않음

섹션 2. 사전 설문 — 기본정보

- **Q(2-1)** 귀하의 연령대는 어떻게 되십니까?
    - ○ 10대
    - ○ 20대
    - ○ 30대
    - ○ 40대
    - ○ 50대 이상
- **Q(2-2)** 귀하의 성별은 무엇입니까?
    - ○ 남성
    - ○ 여성
    - ○ 기타
    - ○ 밝히지 않음
- **Q(2-3)** 온라인에서 패션 상품을 구매하는 빈도는 어느 정도입니까?
    - ○ 거의 없음
    - ○ 연 1~2회
    - ○ 월 1회 미만
    - ○ 월 1~3회
    - ○ 월 4회 이상
- **Q(2-4)** 패션 추천 서비스나 AI 쇼핑 도우미를 몇 회 사용해 보셨습니까? (예: 에이클로젯, Chat-GPT 등)
    - ○ 사용해본 적 없음(0회)
    - ○ 1~2회
    - ○ 3~5회
    - ○ 6~9회
    - ○ 10회 이상
- **Q(2-5)** 패션 상품이나 코디를 고르는 일에 얼마나 익숙하십니까? (1 전혀 익숙하지 않다 ~ 5 매우 익숙하다)

섹션 3. 사전 설문 — 패션 지식과 선호 표현

5점 척도 (1 전혀 그렇지 않다 ~ 5 매우 그렇다)

- **Q(3-1)** 나는 평소 내가 어떤 스타일을 좋아하는지 잘 알고 있다.
- **Q(3-2)** 나는 좋아하는 옷의 스타일을 구체적으로 설명할 수 있다.
- **Q(3-3)** 나는 온라인 쇼핑 시 원하는 옷을 찾을 때 적절한 검색어나 스타일 용어를 사용할 수 있다.
- **Q(3-4)** 나는 마음에 들지 않는 옷&스타일에 대한 이유를 자세히 설명할 수 있다.
- **Q(3-5)** 텍스트 검색이나 말로 표현하기 보단 이미지를 보면서 고를 때 내 취향을 판단하기 쉽다.
- **Q(3-6)** 쇼핑 시 여러 옷을 비교하다가 새로운 주요 쇼핑 기준을 발견한 적이 있다.
- **Q(3-7)** 나는 요즘 유행하는 패션 스타일을 적극적으로 찾아보는 편이다.
- **Q(3-8)** 나는 옷을 고를 때 유행하는 스타일을 참고하는 편이다.
- **Q(3-9)** 나는 유행보다 나만의 스타일을 우선하는 편이다. (역코딩)
- **Q(3-10)** 나는 유행하는 스타일과 나의 스타일 사이의 조화를 고민하는 편이다.
- **Q(3-11)** 나는 유행과 나의 스타일 사이의 조화를 빠르게 파악하는 데 어려움이 있다.

섹션 4. 사전 설문 — 상황에 따른 선택

5점 척도 (1 전혀 그렇지 않다 ~ 5 매우 그렇다)

- **Q(4-1)** 나는 옷을 고를 때 상황에 맞는 스타일을 고려한다.
- **Q(4-2)** 나는 상황에 따른 옷의 스타일을 구체적으로 구분하고 고를 수 있다.
- **Q(4-3)** 중요한 일정에서는 평소 취향과 상황에 맞는 옷 사이에서 고민한다. (예: 소개팅에 어울리는 단정한 캐주얼 스타일과 평소 좋아하는 스트릿 스타일 사이의 고민)
- **Q(4-4)** 상황에 맞추기 위해 평소 스타일을 어느 정도 바꿔야 할지 판단하기 어렵다.
- **Q(4-5)** 온라인 쇼핑 시 특정 상황을 위한 검색 기록이 이후의 평소 추천에 계속 반영되는 것은 원하지 않는다. (예: 면접을 위한 클래식한 정장 스타일 검색기록이 이후 추천에 반영되는 것)

섹션 5. 사전 설문 — AI 패션 추천 도우미가 필요한 상황

- **Q(5-1)** 옷을 고를 때 AI 패션 추천 도우미의 도움이 필요하다고 느끼는 경우를 모두 선택해 주세요. (복수 선택)
    - 내가 원하는 스타일 자체가 아직 명확하지 않을 때
    - 원하는 느낌은 있지만 적절한 패션 용어나 검색어로 표현하기 어려울 때
    - 마음에 드는 스타일의 옷들이 몇가지 있으나 그 이유를 말로 설명하기 어려울 때
    - 평소 내가 좋아하는 스타일을 더 구체적으로 알아보고 싶을 때
    - 평소 취향은 알지만 지금 필요한 조건에 맞춰 어떻게 조정할지 어려울 때
    - 여러 옷 중 무엇을 기준으로 비교하고 선택할지 어려울 때
    - 기존 추천이 내 취향을 충분히 반영하지 못해 개인화 된 맞춤 추천이 필요할 때
    - 새로운 스타일을 시도하고 싶지만 어디서 시작할지 모를 때
    - 유행하는 스타일을 알고 싶거나 이를 내 평소 스타일에 어떻게 녹여낼지 고민될 때
    - 기타
- **Q(5-2)** (선택) 위에서 선택한 사항의 경험을 하나를 떠올려 주세요. 당시 무엇이 막연하거나 어려웠고, 에이전트가 어떤 도움을 주길 바랐는지 구체적으로 적어 주세요. (자유응답)

섹션 6. 기존 시스템(Baseline) — 최종 코디와 의도 표현

5점 척도 (1 전혀 그렇지 않다 ~ 5 매우 그렇다)

- **Q(6-1)** 최종 코디에는 평소 내가 선호하는 스타일이 반영되었다.
- **Q(6-2)** 최종 코디는 주어진 상황에 적합했다.
- **Q(6-3)** 코디를 고르는 과정과 결과에 나는 적극적으로 개입할 수 있었다.
- **Q(6-4)** 시스템을 사용하며 점차 내가 원하는 스타일이 구체화되었다.
- **Q(6-5)** 시스템이 현재 내가 원하는 스타일을 이해했다고 느꼈다.
- **Q(6-6)** 추천에 어떤 정보가 반영되는지 알 수 있었고, 원하는 방향으로 고칠 수 있었다.
- **Q(6-7)** 원하는 요구을 전달하기 위해 정확한 패션 용어를 찾거나 길게 설명할 필요가 없었다.
- **Q(6-8)** 과정을 거치며 옷을 고르는 나만의 기준(색, 핏, 분위기 등)이 분명해졌다.
- **Q(6-9)** 시스템이 보여준 추천은 내가 전달한 의도를 정확히 반영했다.

섹션 7. 제안 시스템(LUMI) — 최종 코디와 의도 표현

5점 척도 (1 전혀 그렇지 않다 ~ 5 매우 그렇다)

- **Q(7-1)** 최종 코디에는 평소 내가 선호하는 스타일이 반영되었다.
- **Q(7-2)** 최종 코디는 주어진 상황에 적합했다.
- **Q(7-3)** 코디를 고르는 과정과 결과에 나는 적극적으로 개입할 수 있었다.
- **Q(7-4)** 시스템을 사용하며 점차 내가 원하는 스타일을 구체화되었다.
- **Q(7-5)** 시스템이 현재 내가 원하는 스타일을 이해했다고 느꼈다.
- **Q(7-6)** 추천에 어떤 정보가 반영되는지 알 수 있었고, 원하는 방향으로 고칠 수 있었다.
- **Q(7-7)** 원하는 요구를 전달하기 위해 정확한 패션 용어를 찾거나 길게 설명할 필요가 없었다.
- **Q(7-8)** 과정을 거치며 옷을 고르는 나만의 기준(색, 핏, 분위기 등)이 분명해졌다.
- **Q(7-9)** 시스템이 보여준 추천은 내가 전달한 의도를 정확히 반영했다.

섹션 8. 제안 시스템 — 전반적 사용성

SUS, 5점 척도 (1 전혀 동의하지 않음 ~ 5 매우 동의함)

- **Q(8-1)** 나는 이 시스템을 자주 사용하고 싶다.
- **Q(8-2)** 이 시스템은 불필요하게 복잡하다고 느꼈다.
- **Q(8-3)** 이 시스템은 사용하기 쉽다고 느꼈다.
- **Q(8-4)** 이 시스템을 사용하려면 전문가의 도움이 필요할 것 같다.
- **Q(8-5)** 이 시스템의 여러 기능은 잘 연결되어 있었다.
- **Q(8-6)** 이 시스템에는 일관성이 부족한 부분이 많았다.
- **Q(8-7)** 대부분의 사람은 이 시스템의 사용법을 빨리 익힐 것 같다.
- **Q(8-8)** 이 시스템을 사용하는 과정은 번거로웠다.
- **Q(8-9)** 나는 이 시스템을 자신 있게 사용할 수 있었다.
- **Q(8-10)** 이 시스템을 사용하기 전에 많은 것을 배워야 했다.

섹션 9. 제안 시스템 — 선호·상황 반영과 상호작용

5점 척도 (1 전혀 그렇지 않다 ~ 5 매우 그렇다)

- **Q(9-1)** 기존 온라인 쇼핑 경험과 비교했을 때, 코디를 고를수록 추천이 이번에 내가 원하는 방향으로 좁혀졌다.
- **Q(9-2)** 시스템이 나의 평소 취향과 상황의 요구를 구분해 반영한다고 느꼈다.

- **Q(9-3)** 이번에 드러난 선호를 평소 취향으로 저장할지 내가 직접 정할 수 있었다.
- **Q(9-4)** 원할 때 언제든 코디를 확정하거나 탐색을 끝낼 수 있었다.
- **Q(9-5)** 말로 설명하기 어려운 취향을 마음에 드는 코디를 고르는 것만으로 전달할 수 있었다.
- **Q(9-6)** 시스템이 제안한 선택 이유(예: "여유로운 핏이 마음에 드셨나요?")는 내가 실제로 고른 이유와 부합했다.. (해당 상황이 없었다면 건너뛰어 주세요)
- **Q(9-7)** 선택 이유를 묻는 질문에 버튼 한 번으로 쉽게 답할 수 있었다. (해당 상황이 없었다면 건너뛰어 주세요)
- **Q(9-8)** 시스템이 내 선택을 어떻게 해석했는지 이해할 수 있었다.
- **Q(9-9)** 시스템이 코디 확정을 제안한 시점이 적절했다. (해당 상황이 없었다면 건너뛰어 주세요)
- **Q(9-10)** 시스템은 꼭 필요할 때만 선택 이유를 물었다.
- **Q(9-11)** 처음 제시된 코디들은 평소 취향과 새로운 스타일을 적절히 섞어 보여주었다.
- **Q(9-12)** 추천 과정에서 평소에 생각하지 못했던 스타일 중 마음에 드는 것을 발견했다.
- **Q(9-13)** 고른 이유를 설명하지 못해도 추천이 막히지 않고 이어졌다. (해당 상황이 없었다면 건너뛰어 주세요)
- **Q(9-14)** 시스템이 내 평소 취향을 기억하고 유지했다고 느꼈다.
- **Q(9-15) (삭제) 상황 제시 요청 뒤에 다른 요청이 없어 측정하지 않으며 폼에서 제외했다. 이후 문항 번호는 그대로 유지한다.**
- **Q(9-16)** 처음 등록한 취향(온보딩)은 평소 내 취향을 잘 나타냈다.
- **Q(9-17)** 이번 과제의 상황에 맞추려면 평소 내 스타일과 다른 선택이 필요했다.
- **Q(9-18)** 저장하기로 한 선호가 두 번째 요청의 추천에 반영되었다. (선호를 저장하지 않았다면 건너뛰어 주세요)
- **Q(9-19)** 이미지 후보를 고르는 과정은 내가 원하는 스타일을 정리하는 데 도움이 되었다.
- **Q(9-20)** 다음 추천을 보면 시스템이 내 이전 선택을 반영했다고 느꼈다.

섹션 10. 마무리 — 자유응답

- **Q(10-1)** 시스템이 자신의 의도에 맞게 작동했다고 느꼈나요? 그렇게 판단한 처음 보여준 5개 코디, 시스템이 제안한 선택 이유, 재추천 결과 등 시스템 사용 경험을 적어 주세요.
- **Q(10-2)** 평소 취향과 특정 상황에서의 요구가 구분되어 작동한다고 느낀 순간이나, 구분되지 않는다고 느낀 순간이 있었나요?
- **Q(10-3)** 제시된 답변 버튼(맞아요 / 다른 부분이 좋아요 등) 중에 맞는 것이 없어서 직접 설명하고 싶었던 순간이 있었다면 적어주세요.
- **Q(10-4)** 비교 이미지가 도움이 됐거나, 더 보여줬으면 했거나, 불필요하다고 느낀 순간이 있었다면 적어주세요.
- **Q(10-5)** AI가 고른 이유를 잘못 짐작했거나, 의도를 잘못 이해했거나, 코디 확정 제안이 너무 빠르거나 늦었다면, 그 순간과 바라는 수정 방식을 적어 주세요.
- **Q(10-6)** 시스템에 대한 전체적인 사용 경험 및 개선사항이 있다면 자유롭게 적어주세요.